CARNOT ENGINE

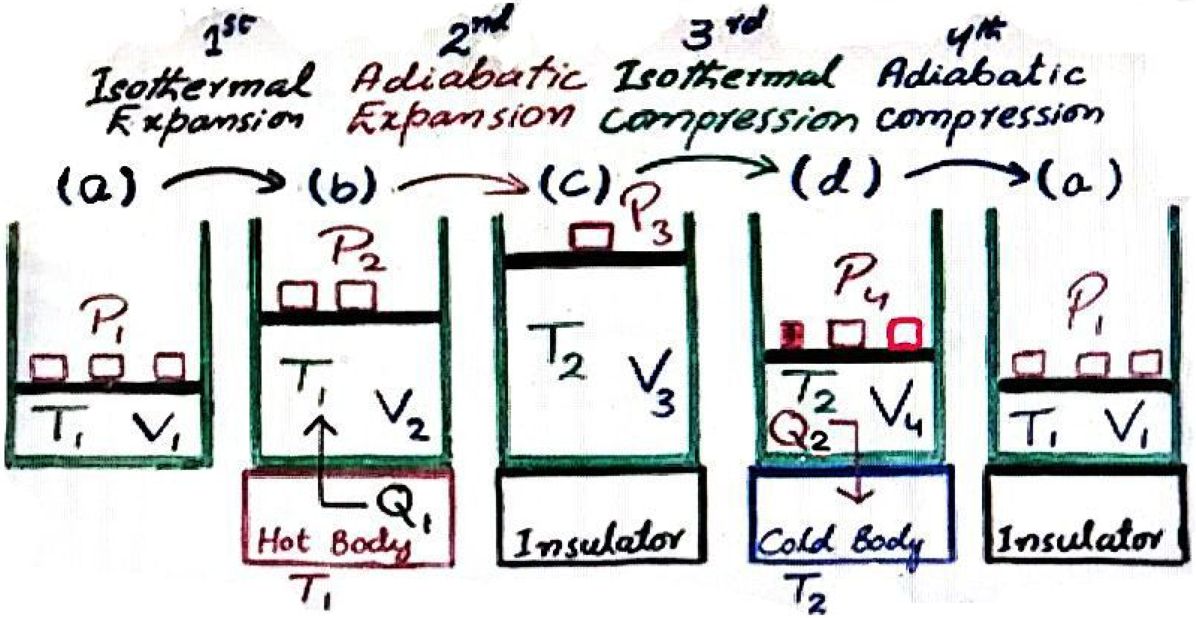
In 1824, a French engineer, ***Sadi Carnot*** proposed the concept of an ideal heat engine has maximum possible efficiency but less than 1 (or 100%).

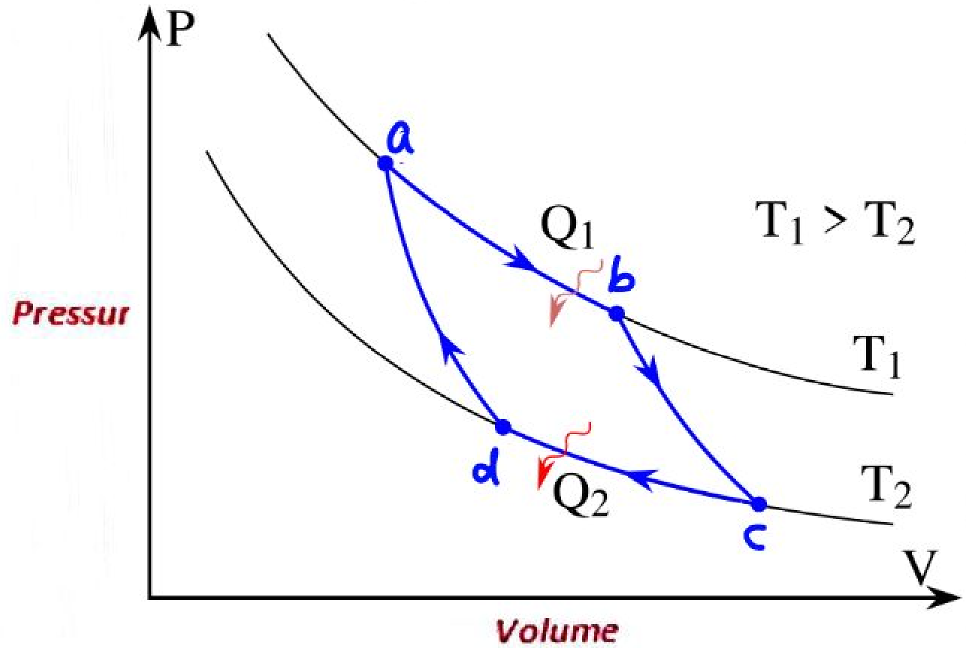
Construction:

It consists of cylinder with insulated walls and moveable piston filled with an ideal gas (working substance), heat can only enter or leave through base.

Carnot cycle:

Operating cycle of the engine is called Carnot Cycle which consists of four steps.





1ST Step: (Isothermal Expansion)

In this step, the cylinder is placed on a hot body (heat source at T_1). Gas is allowed to expand by gradually decreasing load, gas pushes the piston, to keep the temperature (T_1) constant Q_1 heat is added through base.

2ND Step: (Adiabatic Expansion)

In this step, the cylinder is placed on heat insulator. Gas is allowed to expand by gradually decreasing load, gas pushes the piston and work is done at the cost of internal energy so temperature falls from T_1 to T_2.

3RD Step: (Isothermal Compression):

In this step, the cylinder is placed on the cold body (heat sink at T_2). Gas is compressed by gradually increasing load, to keep the temperature (T_2) constant Q_2 heat is rejected to the cold body.

4TH Step: (Adiabatic Compression):

In this step, the cylinder is placed on heat insulator. Gas is compressed to initial state by gradually increasing load, because of work done on the gas internal energy increases so temperature rises from T_2 to T_1.

Work done by the Carnot engine during one cycle is the difference of Q_1 and Q_2

i.e *Work,* $\Delta W = Q_1 - Q_2$ = *(Area bounded by two isotherms and two adiabates)*

Efficiency:

Efficiency is the ratio of output to the input.

$$\boxed{\eta = \frac{Output}{Input} = \frac{Workdone}{HeatAbsorbed} = \frac{\Delta W}{Q_1} 100\%} \rightarrow (i)$$

Since, $\Delta W = Q_1 - Q_2$

$$\eta = \frac{Q_1 - Q_2}{Q_1} 100\% = \left(\frac{Q_1}{Q_1} - \frac{Q_2}{Q_1}\right) 100\%$$

$$\boxed{\eta = \left(1 - \frac{Q_2}{Q_1}\right) 100\%} \rightarrow (ii)$$

According to Carnot's equation,

$\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$, substituting it in eq (*ii*)

$$\left|\eta = \left(1 - \frac{T_2}{T_1}\right) 100\%\right| \rightarrow (iii)$$